1. Who proposed the Chemiosmotic Hypothesis for ATP synthesis in 1961?

(A) Peter Mitchell
(B) Hans Krebs
(C) Fritz Lipmann
(D) Melvin Calvin

2. Which organelle is universally termed the powerhouse of the eukaryotic cell?

(A) Chloroplast
(B) Mitochondria
(C) Peroxisome
(D) Lysosome

3. The inner mitochondrial membrane forms deep invaginations called:

(A) Thylakoids
(B) Cristae
(C) Grana
(D) Stroma

4. Where do the enzymes of the citric acid (TCA) cycle primarily reside?

(A) Cytoplasm
(B) Mitochondrial matrix
(C) Intermembrane space
(D) Outer membrane

5. Which TCA cycle enzyme is embedded in the inner mitochondrial membrane as Complex II?

(A) Citrate synthase
(B) Succinate dehydrogenase
(C) Malate dehydrogenase
(D) Isocitrate dehydrogenase

6. How many net ATP molecules are produced per glucose molecule during glycolysis?

(A) 1 ATP
(B) 2 ATP
(C) 4 ATP
(D) 32 ATP

7. In glycolysis, conversion of phosphoenolpyruvate to pyruvate is catalyzed by:

(A) Hexokinase
(B) Phosphofructokinase-1
(C) Pyruvate kinase
(D) Aldolase

8. Which mobile electron carrier shuttles electrons between Complex III and Complex IV?

(A) Ubiquinone
(B) Cytochrome c
(C) Ferredoxin
(D) Plastoquinone

9. Ubiquinone (Coenzyme Q) is a lipophilic electron carrier dissolved in the:

(A) Mitochondrial matrix
(B) Inner mitochondrial membrane lipid bilayer
(C) Intermembrane space
(D) Outer membrane

10. The terminal electron acceptor of the mitochondrial respiratory chain is:

(A) NAD+
(B) FAD
(C) Molecular oxygen (O2)
(D) Cytochrome c

11. Which respiratory enzyme complex catalyzes the reduction of molecular oxygen to water?

(A) Complex I
(B) Complex II
(C) Complex III
(D) Complex IV (Cytochrome c oxidase)

12. Which poison binds irreversibly to heme a3 in Complex IV, blocking electron transfer?

(A) Rotenone
(B) Cyanide (CN-)
(C) Oligomycin
(D) Malonate

13. Carbon monoxide (CO) exerts toxicity primarily by competing with oxygen for binding to:

(A) Complex I
(B) Complex III
(C) Hemoglobin and Cytochrome c oxidase
(D) ATP synthase

14. Rotenone specifically inhibits electron transfer in which respiratory complex?

(A) Complex I (NADH dehydrogenase)
(B) Complex II
(C) Complex III
(D) Complex IV

15. Antimycin A blocks electron transfer from cytochrome b to cytochrome c1 in:

(A) Complex I
(B) Complex II
(C) Complex III (Cytochrome bc1)
(D) Complex IV

16. Oligomycin inhibits ATP synthesis by directly blocking proton flow through:

(A) F0 subunit of ATP synthase
(B) F1 catalytic head
(C) Complex I
(D) Adenine nucleotide translocase

17. Chemical uncouplers like 2,4-dinitrophenol (2,4-DNP) dissipate the proton gradient as:

(A) ATP
(B) Heat
(C) Light
(D) NADH

18. Thermogenin (UCP1) is a physiological uncoupling protein abundantly expressed in:

(A) White adipose tissue
(B) Brown adipose tissue (BAT)
(C) Skeletal muscle
(D) Liver

19. The light reactions of photosynthesis occur within which chloroplast compartment?

(A) Stroma
(B) Thylakoid membrane and lumen
(C) Outer membrane
(D) Intermembrane space

20. The photosynthetic reaction center of Photosystem II has an absorption peak at:

(A) 680 nm (P680)
(B) 700 nm (P700)
(C) 540 nm
(D) 450 nm

21. The reaction center chlorophyll of Photosystem I has an absorption maximum at:

(A) P680
(B) P700
(C) P870
(D) P430

22. Photolysis of water (water splitting) takes place on the luminal face of:

(A) Photosystem I
(B) Photosystem II (Oxygen-Evolving Complex)
(C) Cytochrome b6f
(D) ATP synthase

23. Which mineral ion is an essential cofactor of the oxygen-evolving complex in PSII?

(A) Manganese (Mn4CaO5 cluster)
(B) Iron
(C) Zinc
(D) Copper

24. The mobile electron carrier that shuttles electrons between PSII and Cytochrome b6f is:

(A) Plastoquinone (PQ)
(B) Plastocyanin (PC)
(C) Ferredoxin
(D) Ubiquinone

25. Plastocyanin is a water-soluble copper-containing protein located in the:

(A) Thylakoid lumen
(B) Chloroplast stroma
(C) Intermembrane space
(D) Inner membrane

26. The enzyme that catalyzes the primary carbon-fixing reaction in the Calvin cycle is:

(A) Phosphoenolpyruvate carboxylase
(B) Rubisco
(C) Carbonic anhydrase
(D) Phosphoglycerate kinase

27. Rubisco fixes atmospheric CO_2 by adding it to which 5-carbon acceptor molecule?

(A) Ribulose-1,5-bisphosphate (RuBP)
(B) Ribose-5-phosphate
(C) 3-Phosphoglycerate
(D) Glyceraldehyde-3-phosphate

28. The primary stable product of CO_2 fixation in C3 plants is:

(A) 3-Phosphoglycerate (3-PGA)
(B) Oxaloacetate
(C) Malate
(D) Glucose-6-phosphate

29. In C4 plants, the primary carbon-fixing enzyme located in mesophyll cells is:

(A) Rubisco
(B) PEP Carboxylase
(C) Pyruvate dehydrogenase
(D) Malic enzyme

30. Kranz anatomy with distinct mesophyll and bundle sheath rings is characteristic of:

(A) C3 plants
(B) C4 plants (e.g., maize, sugarcane)
(C) CAM plants
(D) Algae

31. CAM plants temporally separate carbon fixation by opening their stomata:

(A) Exclusively at night
(B) Exclusively at midday
(C) Continuously
(D) During rain

32. Photorespiration occurs when Rubisco catalyzes the oxygenation of RuBP to produce:

(A) Two molecules of 3-PGA
(B) One 3-PGA and one 2-Phosphoglycolate
(C) Oxaloacetate and CO2
(D) Two glyceraldehydes

33. Photorespiration involves the cooperative metabolic participation of which three organelles?

(A) Chloroplast, Peroxisome, Mitochondria
(B) Chloroplast, Golgi, ER
(C) Mitochondria, Lysosome, Vacuole
(D) Peroxisome, Lysosome, Nucleus

34. Peroxisomes contain which key enzyme to detoxify hydrogen peroxide (H2O2)?

(A) Catalase
(B) Superoxide dismutase
(C) Peroxidase
(D) Glutathione reductase

35. Beta-oxidation of very long-chain fatty acids (> C22) is initiated in:

(A) Mitochondria
(B) Peroxisomes
(C) Smooth ER
(D) Cytoplasm

36. The Glyoxylate cycle in germinating plant seeds converts stored lipids into:

(A) Proteins
(B) Carbohydrates (succinate/glucose)
(C) Nucleic acids
(D) ATP alone

37. Which two unique enzymes define the glyoxylate cycle bypass?

(A) Isocitrate lyase and Malate synthase
(B) Citrate synthase and Fumarase
(C) Aconitase and Dehydrogenase
(D) Pyruvate carboxylase and Kinase

38. The outer mitochondrial membrane is permeable to small molecules (< 5 kDa) due to:

(A) Aquaporins
(B) Porin (VDAC)
(C) TIM23 complex
(D) TOM40 alone

39. Mitochondrial DNA (mtDNA) is maternally inherited in humans and exists as:

(A) Linear chromosomes with histones
(B) Circular, double-stranded molecules lacking histones
(C) Single-stranded RNA
(D) Triplex DNA

40. Cardiolipin constitutes approximately what percentage of total inner mitochondrial membrane lipids?

(A) 1-2%
(B) ~20%
(C) 50%
(D) 80%

41. The standard redox potential (E0') for the NAD+/NADH redox couple is:

(A) -0.320 V
(B) +0.816 V
(C) +0.030 V
(D) 0.000 V

42. The standard redox potential (E0') for the 0.5 O2 / H2O redox couple is:

(A) -0.320 V
(B) +0.816 V
(C) +0.220 V
(D) +0.045 V

43. The total change in standard reduction potential Delta E0' for electron transfer from NADH to O2 is:

(A) +0.496 V
(B) +1.136 V
(C) -1.136 V
(D) +0.816 V

44. Using Delta G0' = -n*F*Delta E0', the standard free energy change for NADH oxidation by O2 (n=2) is:

(A) -219.2 kJ/mol
(B) -50.2 kJ/mol
(C) -30.5 kJ/mol
(D) +219.2 kJ/mol

45. Complex I (NADH:ubiquinone oxidoreductase) pumps how many protons per 2 electrons transferred?

(A) 2 H+
(B) 4 H+
(C) 6 H+
(D) 8 H+

46. Complex II (Succinate dehydrogenase) pumps how many protons across the inner membrane?

(A) 0 H+ (no proton pumping)
(B) 2 H+
(C) 4 H+
(D) 1 H+

47. Complex III (Cytochrome bc1) pumps net protons per pair of electrons via the Q-cycle equal to:

(A) 2 H+
(B) 4 H+
(C) 6 H+
(D) 0 H+

48. Complex IV (Cytochrome c oxidase) translocates how many protons across the inner membrane per 2 electrons?

(A) 1 H+
(B) 2 H+
(C) 4 H+
(D) 0 H+

49. The total number of protons pumped across the inner membrane per pair of electrons from NADH is:

(A) 6 H+
(B) 10 H+ (4 from CI, 4 from CIII, 2 from CIV)
(C) 12 H+
(D) 4 H+

50. The total number of protons pumped per pair of electrons entering via FADH2 (Complex II) is:

(A) 10 H+
(B) 6 H+ (4 from CIII, 2 from CIV)
(C) 4 H+
(D) 2 H+

51. The accepted H+/ATP ratio for the human ATP synthase catalytic cycle (c-ring + Pi symport) is approximately:

(A) 2 H+ per ATP
(B) 4 H+ per ATP (2.7 c-ring + 1 Pi symport)
(C) 6 H+ per ATP
(D) 1 H+ per ATP

52. The P/O ratio (ATP synthesized per oxygen atom reduced) for NADH and FADH2 are approximately:

(A) 2.5 and 1.5 respectively
(B) 3.0 and 2.0
(C) 1.0 and 1.0
(D) 4.0 and 3.0

53. The electrical membrane potential (Delta Psi) across the inner mitochondrial membrane is approximately:

(A) -70 mV
(B) -150 to -180 mV (inside negative)
(C) +60 mV
(D) 0 mV

54. The pH gradient (Delta pH) across the inner mitochondrial membrane is typically:

(A) 0.5 to 1.0 pH units (matrix more alkaline)
(B) 3.0 pH units
(C) 5.0 pH units
(D) 0.0 pH units

55. The Q-cycle mechanism in Complex III allows the transfer of electrons from a two-electron carrier (QH2) to a:

(A) One-electron carrier (Cytochrome c)
(B) Two-electron carrier (O2)
(C) Four-electron carrier
(D) Proton carrier

56. During the Q-cycle, oxidation of two molecules of QH2 results in the net regeneration of:

(A) One QH2 molecule
(B) Two QH2 molecules
(C) Zero QH2
(D) Four QH2

57. Adenine Nucleotide Translocase (ANT) is an antiporter that exchanges:

(A) Matrix ATP4- for cytosolic ADP3-
(B) Matrix ADP for cytosolic ATP
(C) Matrix AMP for cytosolic ATP
(D) Protons for ATP

58. Because ANT exports ATP4- in exchange for ADP3-, its transport is driven electrogenically by:

(A) The membrane potential Delta Psi
(B) Delta pH alone
(C) ATP hydrolysis
(D) Calcium gradient

59. Phosphate translocase imports inorganic phosphate (H2PO4-) into the matrix electroneutrally via:

(A) Symport with H+ (or antiport with OH-)
(B) Antiport with ATP
(C) Facilitated diffusion
(D) Sodium cotransport

60. Bongkrekic acid and atractyloside are specific pharmacological inhibitors of:

(A) Adenine Nucleotide Translocase (ANT)
(B) Phosphate translocase
(C) Complex I
(D) ATP synthase

61. The Glycerol-3-Phosphate shuttle transfers cytoplasmic NADH reducing equivalents into the ETC via:

(A) Mitochondrial glycerol-3-phosphate dehydrogenase reducing FAD to FADH2
(B) Complex I directly
(C) Malate dehydrogenase
(D) Cytochrome c

62. The Malate-Aspartate shuttle yields how many ATP per cytosolic NADH in liver and heart mitochondria?

(A) 1.5 ATP
(B) 2.5 ATP
(C) 1.0 ATP
(D) 3.5 ATP

63. Non-cyclic photophosphorylation produces which two energy-rich products used in the Calvin cycle?

(A) ATP and NADPH
(B) ATP and NADH
(C) GTP and FADH2
(D) ATP alone

64. Cyclic electron flow around Photosystem I generates:

(A) ATP exclusively (no NADPH and no O2)
(B) NADPH exclusively
(C) Both ATP and NADPH
(D) O2 exclusively

65. In cyclic photophosphorylation, electrons from ferredoxin are transferred back to:

(A) Cytochrome b6f complex
(B) Photosystem II
(C) Plastoquinone
(D) Oxygen

66. DCMU (Diuron) is an herbicide that blocks photosynthesis by competing with plastoquinone for the:

(A) QB binding site on Photosystem II
(B) Photosystem I reaction center
(C) Rubisco active site
(D) ATP synthase CF0

67. Paraquat (Methyl viologen) acts as an herbicide by accepting electrons from PSI and generating toxic:

(A) Superoxide free radicals (O2-)
(B) Hydroxyl ions
(C) Singlet nitrogen
(D) Formaldehyde

68. The primary oxygenase product of Rubisco, 2-phosphoglycolate, is dephosphorylated to glycolate in the:

(A) Chloroplast stroma
(B) Peroxisome
(C) Mitochondria
(D) Cytoplasm

69. In photorespiration, conversion of two molecules of glycine to one serine occurs in the:

(A) Mitochondria with release of CO2 and NH3
(B) Peroxisome
(C) Chloroplast
(D) Vacuole

70. The enzyme that detoxifies superoxide radicals (O2-) by converting them into H2O2 and O2 is:

(A) Superoxide Dismutase (SOD)
(B) Catalase
(C) Glutathione Peroxidase
(D) Cytochrome oxidase

71. Glutathione Peroxidase utilizes which trace micronutrient in its catalytic active site (selenocysteine)?

(A) Selenium
(B) Molybdenum
(C) Cobalt
(D) Nickel

72. Mitochondrial uncoupling by UCP1 in brown adipose tissue is allosterically activated by:

(A) Free fatty acids
(B) Purine nucleotides (GDP/ATP)
(C) High pH
(D) Calcium

73. Purine nucleotides (GDP and ADP) act on UCP1 as:

(A) Allosteric inhibitors
(B) Activators
(C) Substrates
(D) Coenzymes

74. Mitochondrial outer membrane permeabilization (MOMP) during apoptosis releases which pro-apoptotic factor?

(A) Cytochrome c
(B) Ubiquinone
(C) Succinate
(D) NADH

75. Cytosolic Cytochrome c binds Apaf-1 in the presence of dATP to form the heptameric:

(A) Apoptosome
(B) Proteasome
(C) Inflammasome
(D) Spliceosome

76. Given Delta Psi = -160 mV and Delta pH = -0.75 units at 310 K, calculate the total proton motive force (pmf) in mV:

(A) -206.5 mV
(B) -160.0 mV
(C) -113.5 mV
(D) -300.2 mV

77. The Mitchell equation for proton motive force is Delta p = Delta Psi - (2.303*R*T / F) * Delta pH. The factor (2.303*R*T / F) at 37 C is:

(A) 59.1 mV
(B) 61.5 mV
(C) 25.7 mV
(D) 96.5 mV

78. If Delta p = -200 mV, the Gibbs free energy change (Delta G) for translocating 1 mole of H+ into the matrix is:

(A) -19.3 kJ/mol
(B) +19.3 kJ/mol
(C) -9.6 kJ/mol
(D) -38.6 kJ/mol

79. If synthesizing 1 mole of ATP requires +50 kJ/mol under intracellular conditions, what is the minimum number of H+ needed at Delta G = -19.3 kJ/mol per H+?

(A) 2.59 protons (approx 3 H+)
(B) 1 proton
(C) 5 protons
(D) 10 protons

80. The c-ring of mammalian mitochondrial ATP synthase consists of how many c-subunits?

(A) 8 subunits (c8 ring)
(B) 10 subunits
(C) 14 subunits
(D) 12 subunits

81. Because the mammalian c-ring contains 8 subunits, one full 360-degree rotation generates 3 ATP by translocating:

(A) 8 protons (yielding 8/3 = 2.67 H+ per ATP)
(B) 12 protons
(C) 6 protons
(D) 3 protons

82. Adding 1 proton per ATP for inorganic phosphate symport, the total H+ cost per ATP in human mitochondria is:

(A) 3.67 H+ per ATP
(B) 2.0 H+ per ATP
(C) 5.0 H+ per ATP
(D) 1.5 H+ per ATP

83. Yeast mitochondrial ATP synthase has a c10 ring, while chloroplast ATP synthase has a c14 ring. Which requires more H+ per ATP?

(A) Chloroplast (14/3 = 4.67 H+ per ATP)
(B) Yeast
(C) Mammals
(D) All require identical H+

84. In the two-half-channel model of the ATP synthase a-subunit, protons move between channels by protonating which conserved residue on the c-subunit?

(A) Aspartate-61 (or Glutamate-58 in yeast)
(B) Lysine-120
(C) Histidine-65
(D) Arginine-210

85. The invariant positively charged residue on subunit a that displaces the proton from the c-subunit is:

(A) Arginine-210
(B) Lysine-45
(C) Aspartate-100
(D) Glutamate-30

86. The Rieske iron-sulfur protein in Complex III and Cytochrome b6f is unique because its [2Fe-2S] cluster is coordinated by:

(A) Two Cysteine and two Histidine residues
(B) Four Cysteines
(C) Four Histidines
(D) Two Methionines

87. The high redox potential of the Rieske iron-sulfur cluster (+280 mV) compared to standard Fe-S clusters (-400 mV) is due to:

(A) Coordination by imidazole nitrogens of two histidines
(B) Absence of iron
(C) Sulfur oxidation
(D) Binding to heme

88. The semi-quinone radical intermediate formed during the Q-cycle at the Qo site can accidentally reduce O2 to generate:

(A) Superoxide radical (O2-)
(B) Hydroperoxide ion
(C) Singlet oxygen
(D) Hydroxyl radical

89. What is the primary site of mitochondrial ROS production under high proton motive force and low ADP (State 4 respiration)?

(A) Complex I via reverse electron transport (RET)
(B) Complex II
(C) Complex IV
(D) ATP synthase

90. In the photo-protection mechanism of plants, Non-Photochemical Quenching (NPQ) dissipates excess excitation energy via:

(A) The Xanthophyll cycle (violaxanthin to antheraxanthin to zeaxanthin)
(B) Calvin cycle acceleration
(C) Rubisco inactivation
(D) Photorespiration arrest

91. The enzyme that catalyzes the de-epoxidation of violaxanthin to zeaxanthin in thylakoids under high light is:

(A) Violaxanthin de-epoxidase (activated by acidic lumen pH < 5.5)
(B) Zeaxanthin epoxidase
(C) Rubisco activase
(D) Superoxide dismutase

92. Rubisco Activase is an AAA+ ATPase that restores Rubisco activity by:

(A) Remotely remodeling the active site to discharge inhibitory sugar phosphates
(B) Phosphorylating Rubisco
(C) Cleaving the small subunit
(D) Binding magnesium

93. Rubisco must undergo carbamylation of which specific active-site lysine residue before binding catalytic Mg2+?

(A) Lysine-201
(B) Lysine-35
(C) Arginine-120
(D) Histidine-50

94. Transition state analog 2-carboxy-D-arabinitol-1,5-bisphosphate (CABP) acts as an extraordinarily tight inhibitor of Rubisco with a Kd of:

(A) ~10^-11 M (picomolar)
(B) 10^-3 M
(C) 10^-6 M
(D) 10^-1 M

95. In C4 photosynthesis, the NADP-malic enzyme subtype decarboxylates malate in the bundle sheath chloroplasts to yield:

(A) Pyruvate, CO2, and NADPH
(B) Oxaloacetate and NADH
(C) Phosphoenolpyruvate and ATP
(D) Citrate

96. The energetic cost of C4 photosynthesis compared to C3 is an extra requirement of how many ATP equivalents per CO2 fixed?

(A) 2 ATP per CO2 fixed (to regenerate PEP via pyruvate phosphate dikinase)
(B) 0 ATP
(C) 5 ATP
(D) 10 ATP

97. Pyruvate Phosphate Dikinase (PPDK) converts pyruvate to PEP utilizing which unique high-energy cleavage?

(A) Cleaves ATP to AMP and PPi
(B) Cleaves ATP to ADP and Pi
(C) Uses GTP
(D) Uses PEP

98. Mitochondrial outer membrane TOM40 complex forms a beta-barrel channel that associates with preprotein receptors:

(A) Tom20 and Tom22
(B) Tim23 and Tim17
(C) Tim50
(D) Sam50

99. Matrix targeting sequences on imported nuclear-encoded mitochondrial precursors typically consist of:

(A) An amphipathic alpha-helix with positively charged Arg/Lys on one face and hydrophobic residues on the other
(B) Hydrophobic beta-sheet
(C) Poly-acidic tail
(D) KDEL motif

100. Translocation of positively charged presequences across the inner membrane via TIM23 is directly driven by:

(A) The electrical membrane potential (Delta Psi, inside negative)
(B) ATP hydrolysis by Tom40
(C) GTPase activity
(D) Delta pH alone

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Formulate Peter Mitchell's Chemiosmotic Hypothesis and explain the two physical components of the Proton Motive Force (pmf).

Q2. Describe the molecular architecture, redox centers, and electron transfer pathway of Complex I (NADH:Ubiquinone Oxidoreductase).

Q3. Explain the Q-Cycle mechanism of Complex III (Cytochrome bc1 Complex) and its role in bifurcated electron transfer.

Q4. Describe the molecular architecture, catalytic centers, and proton translocation mechanism of Complex IV (Cytochrome c Oxidase).

Q5. Detail Boyer's Binding Change Mechanism of F0F1 ATP Synthase and explain how gamma-shaft rotation drives ATP synthesis.

Q6. Explain the biophysical operation of the two-half-channel model of the F0 sector and calculate the proton cost per ATP in mammals.

Q7. Contrast the molecular mechanisms of respiratory uncouplers (2,4-DNP, FCCP) and ATP synthase inhibitors (Oligomycin) on mitochondrial respiration.

Q8. Explain the role of Uncoupling Protein 1 (UCP1 / Thermogenin) in brown adipose tissue non-shivering thermogenesis.

Q9. Describe the Z-scheme of Photosynthesis, detailing the electron flow from water to NADP+ through PSII, Cytochrome b6f, and PSI.

Q10. Explain the catalytic mechanism and Oxygenase-Carboxylase duality of Rubisco, detailing the metabolic pathway of Photorespiration (C2 Cycle).

Q11. Compare C3, C4, and CAM photosynthetic adaptations regarding anatomy, initial carbon fixation, water use efficiency, and temperature response.

Q12. Describe the Glyoxylate Cycle in plant seeds and explain how it enables gluconeogenesis from stored triacylglycerols.

Q13. Explain the biophysical and chemical mechanisms of Reactive Oxygen Species (ROS) generation by the respiratory chain and their enzymatic detoxification.

Q14. Discuss the structural organization and endosymbiotic origins of Mitochondria and Chloroplasts, presenting modern phylogenetic evidence.

Q15. Detail the protein import machinery of the Outer and Inner Mitochondrial Membranes (TOM, TIM23, TIM22, and SAM complexes).

Q16. Explain the biochemical mechanisms of Non-Photochemical Quenching (NPQ) and the Xanthophyll Cycle in photoprotection.

Q17. Describe the architecture and enzymatic regulation of the Pyruvate Dehydrogenase (PDH) Complex, detailing all cofactors and covalent modifications.

Q18. Explain the Glycerol-3-Phosphate and Malate-Aspartate shuttles for translocating cytosolic reducing equivalents into mitochondria.

Q19. Discuss the molecular mechanism of Mitochondrial Permeability Transition Pore (mPTP) opening and its role in ischemic-reperfusion injury.

Q20. Detail the genetic organization of Human Mitochondrial DNA (mtDNA), its transcription mechanism, and maternal inheritance pattern.

Q21. Explain the structural mechanics of the Kok S-State Clock model of photosynthetic water oxidation in the Oxygen-Evolving Complex (OEC).

Q22. Describe the molecular architecture of the Chloroplast ATP Synthase (CF0CF1) and contrast its regulation with the mitochondrial enzyme.

Q23. Explain the metabolic pathway and enzymatic regulation of the Pentose Phosphate Pathway (PPP) in producing NADPH for biosynthesis.

Q24. Discuss the biochemical mechanisms of Cytochrome c release during Apoptosis and the assembly of the Apoptosome.

Q25. Explain how the Malate-Aspartate Shuttle maintains cytosolic redox balance during rapid aerobic glycolysis.

Q26. Describe the molecular mechanisms of Mitochondrial Dynamics (Fusion and Fission) and explain the roles of Mfn1, Mfn2, Opa1, and Drp1.

Q27. Explain the biochemical basis of Cyanide, Carbon Monoxide, and Azide toxicity on Cytochrome c Oxidase.

Q28. Describe the structural organization and function of the Peroxisomal Protein Import Machinery (PEX genes and PTS1/PTS2 signals).

Q29. Explain how the ATP/ADP ratio and energy charge govern coordinate allosteric regulation of Glycolysis and the Citric Acid Cycle.

Q30. Describe the structure and catalytic cycle of the Xanthophyll De-Epoxidase (VDE) enzyme and its dependence on Ascorbate and Thylakoid Lumen pH.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[B]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[B]	**Q62:**	[B]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[A]	**Q43:**	[B]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[B]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[B]	**Q25:**	[A]	**Q45:**	[B]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[C]	**Q27:**	[A]	**Q47:**	[B]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[A]	**Q48:**	[B]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[C]	**Q30:**	[B]	**Q50:**	[B]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[D]	**Q31:**	[A]	**Q51:**	[B]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[C]	**Q33:**	[A]	**Q53:**	[B]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[C]	**Q35:**	[B]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[B]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[B]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[B]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- N2 transfers two electrons to bound Ubiquinone (Q) in the catalytic pocket at the arm interface, reducing it to ubiquinol (QH2).

3. Long-Range Conformational Proton Pumping:

- Complex I pumps 4 H+ across the membrane for every 2 electrons transferred.
- Reduction of Q induces negative charge accumulation, triggering electrostatic rearrangements that drive a piston-like horizontal alpha-helix (HL helix) spanning the entire length of the membrane arm.
- Movement of this central coupling rod coordinates the simultaneous tilting of discontinuous transmembrane helices in ND1, ND2, ND4, and ND5, opening four proton-translocating channels and pumping 4 H+ from matrix to IMS.

>>Solution Question Bank Solutions (Cytochrome c Oxidase).

periplasmic domain of Subunit II.
- Heme a: A low-spin six-coordinate heme located in Subunit I.
- Binuclear Catalytic Center (Heme a3 - Cu_B): Located in Subunit I, forming the active oxygen-binding pocket.
2. Sequential Four-Electron Transfer Pathway:
- Four consecutive molecules of reduced Cytochrome c bind to the IMS face of Subunit II, delivering one electron at a time:
Cytochrome c -> Cu_A -> Heme a -> Heme a3 - Cu_B center.
- In the fully reduced state [Fe(II) - Cu(I)], molecular oxygen (O2) binds between Heme a3 and Cu_B.
- Rapid electron transfer forms a peroxy bridge [Fe(III)-O-O-Cu(II)], which is homolytically cleaved without releasing toxic reactive oxygen intermediates.
- Addition of four chemical protons (4 H+_matrix) reduces the two bound oxygen atoms completely to two molecules of water (2 H2O).
3. Vectorial Proton Pumping:
- In addition to consuming 4 chemical protons from the matrix to make 2 H2O, Complex IV pumps four additional protons (4 H+_pumped) across the membrane through two proton channels (the D-channel and K-channel):
4 Cyt c(red) + 8 H+_matrix + O2 -> 4 Cyt c(ox) + 2 H2O + 4 H+_IMS.
- Yields a net translocation of 2 pumped protons per pair of electrons.

>>Solution to Question 5: Detail Boyer's Binding Change Mechanism of F0F1 ATP Synthase and explain how gamma-shaft rotation drives ATP synthesis.

- ATP synthesis stops completely because no proton gradient remains to drive F0 rotation.
- The free energy of electron transport is entirely dissipated as heat (hyperthermia).

2. ATP Synthase Inhibitors (e.g., Oligomycin):
- Mechanism: A macrolide antibiotic that binds specifically to the contact interface between subunit a and the c-ring of the F0 sector.
- Action: Physically plugs the proton translocation channel, mechanically preventing c-ring rotation and proton re-entry into the matrix.
- Effects on Bioenergetics:
- ATP synthesis is directly and completely blocked.
- Protons pumped by ETC accumulate in the IMS without release; the proton motive force rises to a high thermodynamic equilibrium ceiling (~ -220 mV).
- This massive proton back-pressure halts further proton pumping by Complexes I, III, and IV, causing respiration (O2 consumption) to drop to near zero.
- Diagnostic Test: Adding an uncoupler (2,4-DNP) to oligomycin-inhibited mitochondria immediately restores maximal O2 consumption, proving the ETC machinery itself is intact.

>>Solution Question Bank Solution Question Bank

= -0.32 V) against a thermodynamic gradient of 1.14 V, driven by the sequential absorption of two photons at two distinct photosystems (PSII and PSI).
2. Step 1 (Photosystem II & Water Splitting):
- A photon (680 nm) excites the reaction center chlorophyll P680 to P680* (E0' = -0.80 V).
- Excited P680* rapidly ejects an electron to Pheophytin (Pheo), becoming the extraordinarily strong oxidant P680+ (E0' = +1.25 V).
- P680+ extracts an electron from a redox-active tyrosine residue (Tyr_Z), which in turn extracts electrons from the Mn4CaO5 Oxygen-Evolving Complex (OEC).
- Four successive oxidation steps (Kok's S-state cycle) split 2 H2O molecules into 4 H+, 4 electrons, and 1 molecule of O2 in the thylakoid lumen.
3. Step 2 (Plastoquinone Pool & Cytochrome b6f):
- From Pheophytin, electrons transfer to tightly bound Plastoquinone A (QA), then to mobile Plastoquinone B (QB).
- Upon receiving 2 electrons and 2 protons from the stroma, QB is reduced to Plastoquinol (PQH2), which enters the thylakoid lipid bilayer.
- PQH2 transfers electrons through the Cytochrome b6f complex via a Q-cycle analogous to mitochondrial Complex III, pumping 4 H+ per 2 electrons into the thylakoid lumen.
4. Step 3 (Plastocyanin to Photosystem I):
- Cytochrome b6f passes electrons to Plastocyanin (PC), a water-soluble copper protein diffusing in the thylakoid lumen.
- PC donates its electron to oxidized P700+ in Photosystem I.
5. Step 4 (Photosystem I to NADP+):
- Absorption of a second photon (700 nm) excites P700 to P700*, ejecting an electron through a chain: A0 (chlorophyll a) -> A1 (phylloquinone / vitamin K1) -> FeS clusters (FX, FA, FB).
- From FB, electrons reduce Ferredoxin (Fd), a soluble stromal [2Fe-2S] protein.
- Ferredoxin-NADP+ Reductase (FNR) catalyzes the transfer of two electrons from two reduced ferredoxin molecules plus a stromal proton to NADP+, yielding NADPH.

>> **Solution to Question 10:** Explain the catalytic mechanism and Oxygenase-Carboxylase duality of Rubisco, detailing the metabolic pathway of Photorespiration (C

hydroxypyruvate, which is reduced to Glycerate. Glycerate enters the chloroplast, where glycerate kinase consumes 1 ATP to phosphorylate it into 3-PGA, re-entering the Calvin cycle.

3. Energetic Cost:

Photorespiration dissipates up to 30-40% of fixed carbon and consumes ATP and reducing equivalents without producing sugar, creating strong evolutionary pressure for the evolution of C4 and CAM carbon-concentrating mechanisms.

---

growth before photosynthetic emergence.
2. The Two Unique Bypass Enzymes:
- Isocitrate Lyase: Cleaves isocitrate (6C) into Succinate (4C) and Glyoxylate (2C), bypassing the two oxidative decarboxylation steps.
- Malate Synthase: Condenses Glyoxylate (2C) with a second molecule of Acetyl-CoA (2C) to form Malate (4C).
3. Pathway Integration Across Compartments:
- Glyoxysome (Specialized Peroxisome): Fatty acids undergo beta-oxidation to generate Acetyl-CoA. Acetyl-CoA condenses with oxaloacetate to form citrate, isomerized to isocitrate. Isocitrate lyase releases succinate. Malate synthase condenses glyoxylate with another Acetyl-CoA to make malate.
- Mitochondria: Succinate exits the glyoxysome, enters the mitochondrial matrix, and is converted by TCA enzymes into Fumarate -> Malate -> Oxaloacetate.
- Cytosol: Malate is exported to the cytosol, oxidized to oxaloacetate, and decarboxylated by Phosphoenolpyruvate Carboxykinase (PEPCK) to PEP, entering reverse glycolysis (gluconeogenesis) to yield sucrose.

>>Solution to Question 13 Explain the biophysical and chemical mechanisms of Reactive Oxygen Species (ROS) generation by the respiratory chain and their en

4. Inner Membrane Insertion via TIM22:
Hydrophobic, multi-pass inner membrane metabolite carriers (e.g., ANT, phosphate translocase) lack cleavable N-terminal presequences. They are escorted across the IMS by Tim9-Tim10 chaperones and inserted into the inner membrane by the carrier translocase TIM22, driven by Delta Psi.

---

2. Five Sequential Chemical Steps and Cofactors:
- Step 1: E1 decarboxylates pyruvate, transferring the 2C hydroxyethyl group to TPP, releasing CO2.
- Step 2: E1 oxidizes the hydroxyethyl group to an acetyl group and transfers it to the disulfide bond of lipoamide on the swinging arm of E2.
- Step 3: E2 transfers the acetyl group from lipoamide to Coenzyme A (CoA-SH), releasing high-energy Acetyl-CoA.
- Step 4: The reduced, dithiol lipoamide arm swings to E3, where it is re-oxidized by E3's disulfide bond, restoring lipoamide.
- Step 5: The dithiol of E3 transfers electrons to FAD, forming FADH2, which in turn reduces NAD+ to NADH + H+.
3. Covalent Phosphorylation/Dephosphorylation Regulation:
- Inactivation: Pyruvate Dehydrogenase Kinase (PDK) phosphorylates three specific serine residues on the E1-alpha subunit (Ser-264, Ser-271, Ser-203), completely inactivating the complex. PDK is allosterically activated by high energy charge: [NADH]/[NAD+], [Acetyl-CoA]/[CoA], and [ATP]/[ADP].
- Activation: Pyruvate Dehydrogenase Phosphatase (PDP) dephosphorylates E1-alpha, restoring full activity. PDP is activated by Ca2+ (signaling muscle contraction) and insulin (signaling energy abundance).

Solution to Question 18

>> Explain the Glycerol-3-Phosphate and Malate-Aspartate shuttles for translocating cytosolic reducing equivalents into mitochondria.

---

and Light-Strand Promoters (HSP and LSP).

3. Polycistronic Transcription:

Mitochondrial RNA polymerase (POLRMT) initiates transcription from HSP and LSP, synthesizing massive polycistronic precursor RNAs spanning almost the entire circular chromosome.

The 'tRNA Punctuation Model': tRNAs interspersed between protein-coding genes act as recognition signals cleaved by mitochondrial RNase P (5'-end) and RNase Z (3'-end) to liberate individual mRNAs and rRNAs.

4. Maternal Inheritance and Replicative Segregation:

Sperm mitochondria (~100 copies) entering the oocyte are tagged with ubiquitin and selectively destroyed by autophagy (mitophagy). Consequently, mtDNA is inherited exclusively through the maternal lineage. Mutations exhibit heteroplasmy (mixture of mutant and wild-type genomes); clinical phenotypes manifest only when mutant load exceeds a critical biochemical threshold (typically 60-80%).

Homologous to mitochondrial ATP synthase, CF0CF1 is localized exclusively in non-appressed thylakoid stroma lamellae and grana margins:
- CF0 (Membrane Domain): Composed of subunits I, II, IV, and a large ring of 14 hydrophobic III (c) subunits (c14 ring).
- CF1 (Stromal Catalytic Head): A hexamer of alternating alpha and beta subunits, (alpha-beta)3, connected to the c14 ring via central rotor stalk gamma and epsilon subunits.
2. Functional Consequence of the Large c14 Ring:
Because the c-ring contains 14 subunits, one 360-degree rotation translocates 14 protons and produces 3 ATP:
- H+/ATP ratio = 14 / 3 = 4.67 H+ per ATP!
- This high stoichiometric proton requirement reflects the large electrochemical pH gradient (Delta pH ~ 3 units, lumen pH ~ 5.0 vs stroma pH ~ 8.0) and virtually zero Delta Psi in steady-state chloroplasts (due to counter-ion Cl- and Mg2+ fluxes).
3. Unique Light-Dependent Redox Regulation:
Unlike mitochondria, which operate continuously, chloroplast ATP synthase must be shut down in the dark to prevent wasteful reverse hydrolysis of stromal ATP:
- The gamma subunit of CF1 possesses a unique 38-amino-acid insertion containing two conserved cysteine residues (Cys-199 and Cys-205).
- In the dark: The cysteines form an intramolecular disulfide bond (S-S), locking the gamma subunit in an inactive conformation.
- In the light: Photosynthetic electron transport reduces Ferredoxin. Ferredoxin-Thioredoxin Reductase (FTR) transfers electrons to Thioredoxin-m, which reduces the CF1-gamma disulfide bond to free sulfhydryls (-SH HS-), unlocking rotation and activating ATP synthesis.

>>**Solution to Question 23:** Explain the metabolic pathway and enzymatic regulation of the Pentose Phosphate Pathway (PPP) in producing NADPH for biosynthes

severe hemolytic anemia.

>>Solution Question 28: Discuss how the Mature Anaphase Stage Complex is released during Anaphase and the assembly of the Apoptosome.

Matrix NADH, which directly donates electrons to Complex I of the respiratory chain.
- Step 4: To return the 4-carbon skeleton across the membrane, matrix OAA is transaminated with Glutamate by Mitochondrial Aspartate Aminotransferase (mAST), producing Aspartate and alpha-Ketoglutarate.
- Step 5: The Glutamate-Aspartate Antiporter (Aralar / SLC25A12) exports Aspartate to the cytosol in exchange for Glutamate.
- Step 6: Cytosolic Aspartate Aminotransferase (cAST) transaminates Aspartate with alpha-ketoglutarate back to OAA and Glutamate, completing the closed cycle.
3. Energetic Advantage:
Unlike the glycerol-3-phosphate shuttle, which loses energy by feeding into ubiquinone, the malate-aspartate shuttle preserves the full high-potential energy of cytosolic NADH, yielding 2.5 ATP per NADH.

>>Solution Question 28: Describe the biochemical basis of Cyanide, Carbon Monoxide, and Azide toxicity. Mfn1, Mfn2, Opa1, and D

In Complex IV, the heme a3 iron atom is five-coordinate, presenting an open axial coordination position designed to bind molecular oxygen (O2) between Fe(II) and Cu_B(I).

2. Mechanism of Cyanide (CN-) Poisoning:

- Cyanide anion (CN-) is an isoelectronic analog of O2 that diffuses rapidly across membranes.
- CN- binds with extraordinarily high affinity (Kd ~ 10^-7 M) to the oxidized ferric [Fe(III)] state of heme a3.
- Coordination of cyanide permanently locks the binuclear center in a rigid, non-functional coordination geometry, completely blocking electron transfer from heme a to heme a3 and preventing oxygen reduction to water.

3. Mechanism of Carbon Monoxide (CO) Poisoning:

- Carbon monoxide binds specifically to the reduced ferrous [Fe(II)] state of heme a3, directly competing with O2.
- While CO's primary toxicity stems from high-affinity binding to hemoglobin (forming carboxyhemoglobin, left-shifting the O2 dissociation curve), its direct inhibition of Complex IV contributes significantly to cellular histotoxic hypoxia in high-metabolic tissues (brain and myocardium).

4. Mechanism of Azide (N3-) Poisoning:

- Azide binds to both Fe(III) of heme a3 and Cu_B, bridging the two metal centers and halting the catalytic cycle identically to cyanide.

5. Physiological Consequences and Antidote Biochemistry:

- Complete inhibition of Complex IV halts electron flow throughout the entire respiratory chain (Complexes I-III become fully reduced).
- Proton pumping ceases; the proton motive force collapses; ATP synthesis stops immediately.
- Cells shift to anaerobic glycolysis, producing massive lactic acidosis and rapid cytotoxic edema.
- Antidote Therapy: Administration of Hydroxocobalamin (Vitamin B12a) binds cyanide with higher affinity than heme a3, forming non-toxic Cyanocobalamin excreted in urine; or Sodium Nitrite, which oxidizes hemoglobin to methemoglobin [Fe(III)], sequestering cyanide away from mitochondria.

Solution to Question 28: >>Describe the structural organization and function of the Peroxisomal Protein Import Machinery (PEX genes and PTS1/PTS2 signals).

4. Receptor Ubiquitination and Recycling:
- Following cargo release, PEX5 is mono-ubiquitinated at a conserved cysteine residue (Cys-11) by the membrane-bound ubiquitin-conjugating enzyme PEX4 (E2) and RING-finger ligases PEX2, PEX10, PEX12 (E3).
- The hexameric AAA+ ATPase complex PEX1-PEX6 utilizes ATP hydrolysis to pull mono-ubiquitinated PEX5 out of the peroxisomal membrane back into the cytosol, where it is de-ubiquitinated and recycled.
5. Clinical Disorder (Zellweger Syndrome):
Mutations in PEX genes (most commonly PEX1) abolish peroxisome biogenesis, leaving empty peroxisomal 'ghosts'. Severe accumulation of very long-chain fatty acids (VLCFAs) causes craniofacial dysmorphism, hypotonia, hepatomegaly, and infant death.

---

the monoepoxide Antheraxanthin, and finally to the epoxide-free Zeaxanthin during photoprotective Non-Photochemical Quenching (NPQ).

2. Domain Organization:

- N-terminal Cysteine-Rich Domain: Contains 13 conserved cysteine residues involved in essential intramolecular disulfide bonds stabilizing protein folding.
- Central Lipocalin Domain: Forms an 8-stranded antiparallel beta-barrel possessing a hydrophobic cavity that accommodates the bulky, hydrophobic beta-ionone ring and polyene tail of the xanthophyll substrate.
- C-terminal Glutamate-Rich Domain: Contains a high density of acidic glutamate and aspartate residues acting as the primary pH-sensor.

3. Strict pH-Dependent Conformational Switch:

- In the dark or low light (lumen pH ~ 7.0 to 7.2): VDE exists as an inactive, soluble monomer freely diffusing in the aqueous thylakoid lumen. Its negatively charged C-terminal glutamate residues cause electrostatic repulsion, preventing membrane binding.
- Under high light (lumen pH drops below 5.5): Protonation of the glutamate residues neutralizes negative surface charges. VDE undergoes dimerization and conformational rearrangement, exposing hydrophobic surface patches that drive reversible electrostatic and hydrophobic binding to the thylakoid lipid bilayer.

4. Cofactor Requirement (Ascorbic Acid):

- In the active membrane-bound dimer state, VDE requires Ascorbic Acid (Vitamin C) as an obligatory co-substrate.
- Ascorbate binds to the active site, serving as the two-electron donor that reduces the epoxide oxygen, liberating it as water and generating dehydroascorbate, which is subsequently regenerated in the stroma by the ascorbate-glutathione cycle.
- The resulting zeaxanthin molecules accumulate in the light-harvesting antenna complexes, directly quenching singlet excited chlorophylls to dissipate excess solar energy safely as heat.
